Accurate Co-Optimization of Crop Light and Electricity by Sensor-Driven Panel Tilting on a Wire-Rope Agrovoltaic Canopy: A Simulation Study at Madurai, India

Author names and affiliations withheld for review

Abstract - Agrovoltaic farming shares land between crops and solar panels, but a fixed panel angle favors either the crop or the electricity. We ask how accurately a controller can serve both at once by tilting the panels from sensor readings. The panels of a stainless-steel wire-rope canopy tilt together, driven by one motor and one drive wire, and a controller sets the angle from six sensors: a pyranometer, a light sensor under the panels, an air temperature and humidity sensor, a soil moisture probe, a rain sensor, and an anemometer. It lays the panels flat in rain, turns them edge-on in storm wind, and otherwise picks the angle with the best equal-weight mix of electricity and crop light. In a MATLAB model and a browser digital twin of a campus plot at Madurai, India, 864 cases with noisy sensor readings gave 99.6 percent of the best possible combined score, with 82 percent of the best electricity and 98 percent of the best crop light value, against 91.2 percent for a fixed 45 degree tilt. The score stayed between 99.5 and 99.7 percent across sky, soil, and season, and with a failed light sensor. All results are simulated.

Index Terms - Agrovoltaics, crop light, sensor-based control, tilt mechanism

INTRODUCTION

Agrovoltaic systems grow crops and generate electricity on the same land. Studies of such systems report or predict a higher combined land productivity than growing the two separately [1]–[4], and some crops hold their yield under partial shade because they intercept light more efficiently [5]. How much the crop gains depends on the light it receives and when it receives it [3]. With fixed panels, that light is set once, at design time. A panel steep enough to let light through to the crop loses electricity, and a panel at the angle for maximum electricity shades the crop heavily.

A tilting panel can move this compromise hour by hour. That needs sensors, a controller, and a structure that can turn many panels together at low cost. This paper describes such a system for a small farm: stainless-steel wire ropes carry rows of panels 4.5 m above the crop, and one motor moves one drive wire to tilt every row. DIN SPEC 91434 [6] lists light availability and light homogeneity under the array among its planning requirements, which is the reason we treat crop light as a design quantity and not as a by-product.

The question of this paper is how accurately a controller can serve both goals at once, using sensors and the panel tilt. The study is a computer study of the Thiagarajar College of Engineering ground in Madurai (9.88° N, 78.08° E). Our contributions are as follows:

- a measure of co-optimization accuracy, and a test of a six-sensor controller against the best possible angle under noisy and faulty sensors and across sky, soil, and season;
- the controller itself, which scores both goals in one objective, corrects its crop light model with a light sensor under the panels, relaxes the crop light need under heat and drought stress, and has a safety layer that handles sensor faults, limits panel movement, and counts motor wear;
- a tilt mechanism with one drive wire and a lever arm fixed at a 45° offset, which keeps a usable lever at the 90° storm position;
- an honest look at where the approach is weak: the rupee value of crop income against electricity, the water saving, and the storm case.

Related Work

Agrophotovoltaic systems have been reviewed in terms of applications, challenges and opportunities [3]. A demonstration plant in Germany reported land-equivalent-ratio gains of 56 to 70 percent in 2017 and close to 90 percent in the hot, dry summer of 2018, with a vertical clearance of 5 m for farm machinery [4]. A modelling study predicted land-productivity gains of 35 to 73 percent, which its authors called preliminary [1], and a field study in drylands reported benefits across food, energy, and water [2]. These studies concentrate on layout and crop response. This paper concentrates on how a tilting array should be controlled, and on what that control is worth compared with fixed and monthly-scheduled tilt. The controller combines five elements: one objective that scores both goals with a crop light model; a light sensor under the panels that corrects that model; humidity and soil moisture that lower the light the crop needs when it is stressed; sensor-fault fallbacks; and limits on how often the panels move. We have not carried out a formal prior-art search, so we make no claim beyond this description that the combination is new.

System Description

I. Canopy geometry

The geometry is held in one design file that feeds the MATLAB engine, the browser digital twin, the drawings, and the 3D model. The checked design has ten rows of panels, each 2.497 m wide, on a 2.5 m pitch along a 25 m run. The pivot axis is 5.58 m above ground, so the lowest panel edge is 4.50 m high at the 60° limit. Carrier ropes of 16 mm stainless steel (SS316, 7x19) are spaced every 5 m and sag by a span/12. The torque tube is a CHS 114.3 x 4 mm section, the 36 columns are RHS 200 x 200 x 6 mm, and the stays are 20 mm ropes. Fig. 1 shows a four-row section.

An earlier layout with 12.5 m bays failed a screening check made with the wind loads of IS 875 (Part 3) [7] and a basic wind speed of 39 m/s, which we still have to confirm against the code map: its rope was loaded 16 times its allowed value and its 60.3 mm tube 30 times. The 5 m grid passes. Table 1 gives the loads and capacities of the final design at 140 km/h wind with the panels held at 60°. This is a hand calculation at screening level; connections, foundations, fatigue and dynamic wind are not designed.

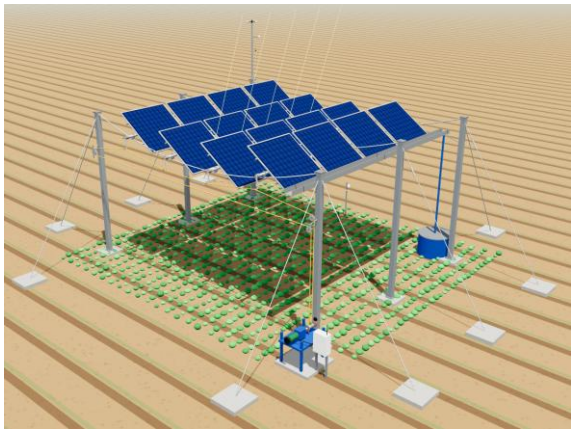


Figure 1. Canopy section in 3D. Four rows of panels hang from carrier ropes between columns. The winch, drum and drive wire are at the front, the weather mast is at the back left, and the gutter and cistern are at the right. Wires and ropes are drawn about four times thicker than real so that they can be seen.

Table 1. Strength check of the final design at 140 km/h wind

Part	Load	Capacity	Ratio
Carrier rope 16 mm	26.8 kN	72.7 kN	0.37
Stay rope 20 mm	36.1 kN	114 kN	0.32
Torque tube 114.3 x 4	10.0 kNm	13.11 kNm	0.76
Bearing block 50 mm	8.5 kN	23 kN	0.37
Column 200 x 200 x 6	8.5 kN	802 kN	0.05
Column pad (uplift)	7.9 kN	11.4 kN	0.69
Drive wire 6 mm*	4.1 kN	10.2 kN	0.41
Winch 10 kN*	4.1 kN	10 kN	0.41

Ratio is load divided by capacity. Capacities are indicative catalog values, so the ratios are screening results, and the rope capacities are half the breaking loads. The entries marked with an asterisk are for a 54 km/h wind with the panels held at 60°.

II. Tilt mechanism

All rows are tilted by one drive wire. The wire is clamped to the tip of a lever arm of length $L = 0.44$ m that is rigidly fixed to the end of each torque tube. The arm is set at an offset of $\alpha = 45°$, so that it points down and to one side when the panel is at its flat position of 3°. When the wire pulls the tip sideways, the tube turns. The wire travel from the flat position to a tilt θ is

$$d(\theta) = L\,[\sin(\theta - \alpha) - \sin(3° - \alpha)], \quad (1)$$

and the useful lever length is

$$l(\theta) = L\cos(\theta - \alpha). \quad (2)$$

With these values the wire moves 408 mm to reach 60° and 606 mm to reach the 90° storm position. A plain arm without the offset has no lever left at 90°, while the offset arm keeps at least 0.31 m (Fig. 2). A drum of 120 mm

diameter turns 1.08 times to reach 60° and 1.61 times to reach 90°. Both ends of the drive wire are fastened to the drum, so winding one strand in pays the other out and the wire pulls in both directions; no return spring is needed.

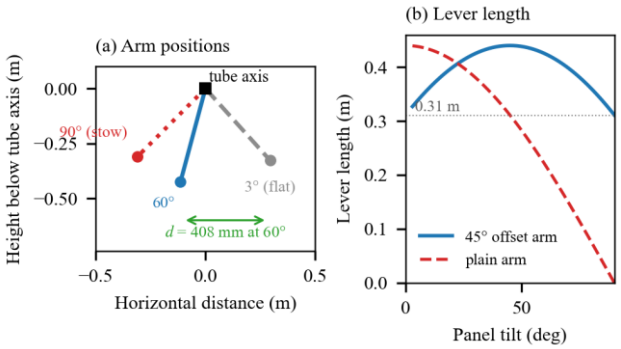


FIGURE 2. LEVER ARM GEOMETRY. (A) ARM POSITIONS AT THE FLAT POSITION, AT 60°, AND AT THE 90° STORM POSITION, WITH THE WIRE TRAVEL TO 60°. (B) LEVER LENGTH AGAINST TILT FOR THE 45° OFFSET ARM AND FOR A PLAIN ARM.

Each row of four panels weighs about 176 kg. The wire force is 0.4 kN in calm air, 2.1 kN at 36 km/h, and 4.1 kN at 54 km/h, using a flat-plate coefficient of 1.0, which is conservative. A 6 mm SS316 drive wire (10.2 kN allowed) and a 10 kN worm-gear winch of about 0.25 kW therefore work at a utilization of 0.41 or less up to 54 km/h. A first parts list with a 3 kN winch and a 4 mm wire was rejected by this check. Going from 15° to 60° needs 0.33 m of wire; in 45 s that is 7.4 mm/s, a mechanical power of about 37 W at 5 kN. At 15 m/s (54 km/h) the controller stows the panels edge-on instead of moving them, as described under Models and Controller.

III. Sensors

Six sensors feed the controller. A pyranometer, a rain sensor, and a cup anemometer sit on a 3 m mast above the canopy. A light sensor for photosynthetically active radiation (PAR), a DHT11 temperature and humidity sensor, and a soil moisture probe are placed under the panels. In this study the sensor values are simulated. A separate prototype reads a DHT11, a soil probe, and a rain sensor with a microcontroller board; the other three are not yet measured.

IV. Prototype stage

A first prototype reads the DHT11, the soil probe, and the rain sensor with a microcontroller board and runs the decision rules in a MATLAB dashboard. Sixteen software-in-the-loop tests all passed. They cover the serial parser, a failed DHT11 read being flagged and not used, soil calibration, the vapor pressure deficit (2.12 kPa at 30 °C and 50 percent humidity), the rain rule (3°), the wind rule (stow at 15.0 m/s but not at 14.9 m/s, with stow overriding rain), night parking, the optimizer matching a brute-force search (43.5°), a 1.9 percent electricity gain over a fixed 45° panel at clear noon, heat stress moving the angle from 43.5° to 34.5°, the pump hysteresis, and agreement of zone light and power with the JavaScript engine. Bench measurements with the board are not reported in this paper. A logging tool that scores a week of readings against the best angle is ready; we rehearsed it on a synthetic week only, which tests the tool and says nothing about the site.

MODELS AND CONTROLLER

I. Solar, light and water models

Sun position follows the usual declination and hour-angle equations for latitude 9.88° N. Monthly climate comes from the NASA POWER data set [8]. Hourly irradiance is the clearness index times the extraterrestrial value, and the diffuse fraction follows Erbs et al. [9]. Panel power uses the Huld et al. model [10] with 14 percent system losses, a row self-shading term, and an incidence-angle modifier. The light on the ground is found by casting rays in the east–west plane to 40 points per panel pitch every hour, and four zone strips per pitch carry four crops: ginger, turmeric, coriander, and amaranth.

Reference evapotranspiration follows the temperature method of Hargreaves and Samani [11], and the daily root-zone water balance follows FAO-56 [12]. Yield is the potential yield times factors for light, temperature, and soil, reduced for water stress by the yield response factor of Doorenbos and Kassam [13]:

$$Y = Y_{\mathrm{pot}} f_{\mathrm{L}} f_{\mathrm{T}} f_{\mathrm{S}} \left[1 - K_{\mathrm{y}} \left(1 - \mathrm{ET_a} / \mathrm{ET_c}\right)\right], \qquad (3)$$

where f_{L}, f_{T}, and f_{S} are the light, temperature, and soil factors, $\mathrm{ET_a}$ and $\mathrm{ET_c}$ are actual and crop evapotranspiration, and K_{y} is the yield response factor. The light factor is 1 at or above an optimum light level for each crop, falls linearly to 0.35 at a minimum level, and falls to zero in darkness. Crop parameters are typical literature-style values rounded for a screening model. Climate, soil, and crop data are listed in the project evidence folder.

II. Six-sensor controller

The controller applies the rules of Table 2 in order. When it optimizes, it tries every angle from 15° to 60° (a coarse step of 5°, then 0.5°) and keeps the angle that maximizes

$$J(\theta) = w\, P(\theta) / P_{\max} + (1 - w)\, C(\theta) / C_{\max}, \qquad (4)$$

where P is panel power, C is the price-weighted light value of the four crop zones, $P_{\max}$ and $C_{\max}$ are their maxima over the angles, and w is the electricity weight, set to 0.5. Crop value counts only in the growing season of each zone.

TABLE 2. CONTROLLER RULES IN ORDER OF PRIORITY

No.	Condition	Action
1	Wind 15 m/s or more (resume below 12 m/s)	Stow at 90°
2	Rain sensor wet	Flat at 3°, water to gutter
3	Pyranometer below 20 W/m²	Night park at 15°
4	Otherwise	Maximize J over 15° to 60°

The PAR sensor corrects the crop light model. The model predicts the ratio of PAR to pyranometer reading at the angle in use, and the correction is

$$c = (\mathrm{PAR} / \mathrm{GHI})_{\mathrm{measured}} / (\mathrm{PAR} / \mathrm{GHI})_{\mathrm{model}}, \quad (5)$$

limited to the range 0.7 to 1.3, where GHI is the global horizontal irradiance. A ratio outside 0.5 to 1.6 is treated as a faulty PAR sensor and ignored. Hot, dry air (a high vapor pressure deficit) and dry soil lower the light the crop needs by up to 30 percent, so the panels give more shade in a way that suits the stressed crop. Each channel is the median of five readings, and a new angle is accepted only if it improves J by more than 2 percent, which stops the panels from hunting.

A safety layer wraps the controller. A soil probe that gives no reading, reads outside 0 to 100 percent, jumps more than 40 points in an hour, or repeats the same raw value for three daytime hours is replaced by the water-balance value or the last good value. A rain sensor that stays wet for 30 minutes in dry, sunny air is ignored, and a dry reading with a rain forecast, humidity of 90 percent or more, and a dark sky counts as rain. Panels must hold for at least 120 s between moves and may move at most 12 times an hour and 900° a day; rain, storm, and night moves are exempt. The layer also counts moves, panel travel, wire length, and drum turns.

VERIFICATION

The panel power model was compared with the PVGIS ERA5 data [14] for a horizontal plane at the site. The annual energy is 5.0 percent higher and the annual irradiation 5.2 percent higher than PVGIS; the monthly difference is between –1.0 and 10.8 percent and is largest in July. The shadow model was compared with a 3D ray tracer (Blender Cycles, direct light only) in 80 cases: the mean absolute difference in lit fraction is 0.010, the largest difference is 0.22, and 96 percent of cases are within 0.10. The JavaScript twin and the MATLAB engine agree to 0.0000 percent on 22 outputs. This shows that they are the same model; the comparisons with PVGIS and the ray tracer test the physics. The crop, water, and economic models are not validated against field data.

The controller was coded twice, in JavaScript for the twin and in MATLAB, with a Simulink version for the supervisory logic. On 40 random sensor states the two codes gave the same mode and the same angle in all 40 cases. The safety layer passed all eight of its tests: a rain sensor stuck wet, a dead rain sensor, an unplugged soil probe, a glitching probe, a stuck probe, an out-of-range probe, cloud flicker, and wear on a clear day.

ACCURACY OF THE CONTROLLER

The reference is the best angle for the same objective (4), found by searching every angle with perfect knowledge of the weather. We scored each strategy hour by hour against it. The test has 864 cases: a representative day for each month, three sky states (clear, partly cloudy, overcast), two soil states (26 and 15 percent moisture), and the daylight hours of each day. Sensors were the true values plus random noise with standard deviations of 5 percent for the pyranometer and PAR sensor, 1.2 °C for temperature, 3 percent for relative humidity, and 1.8 points for soil moisture. The co-optimization accuracy of a strategy is

$$A = 100\, \Sigma_k\, g_k\, J(\theta_k) \,/\, \Sigma_k\, g_k\, J(\theta_k^{*}), \quad (6)$$

where θ_k is the angle the strategy chooses in case k, θ_k^{*} is the best angle, and g_k is the case weight times the global horizontal irradiance of the case. The crop light score uses the same sum with $C/C_{\max}$ in place of J, and the electricity score is the electricity produced divided by the best possible, summed with the case weights. A is 100 when the strategy always picks the best angle.

TABLE 3. ACCURACY AGAINST THE BEST POSSIBLE ANGLE (864 CASES)

Strategy	Energy (%)	Crop (%)	Score (%)	Error (deg)
Controller, ideal sensors	81.9	98.1	99.7	5.2
Controller, noisy sensors	82.1	97.7	99.6	4.8
Controller, single readings	81.8	97.5	99.4	5.5
Monthly plan	82.2	96.3	98.8	9.5
Fixed 45°	81.0	84.6	91.2	16.3
Fixed 15° (electricity only)	99.2	45.2	80.0	34.0
Sun tracking	88.6	64.3	84.0	25.0

Table 3 and Fig. 3 give the result. In the table, Energy is electricity and Crop is the crop light value, each as a percentage of its best possible value, Score is the combined score, and Error is the root-mean-square tilt error. With noisy sensors the controller reaches 99.6 percent of the best combined score, with a tilt error of 4.8° root mean square. It gets 82 percent of the best electricity because the best electricity belongs to a low angle that would shade the crop, and it gets 98 percent of the best crop light value. A fixed 15° panel gets 99 percent of the electricity but only 45 percent of the crop light value. The monthly plan, which is the schedule the twin uses without live sensors, scores 98.8 percent with a tilt error of 9.5°.

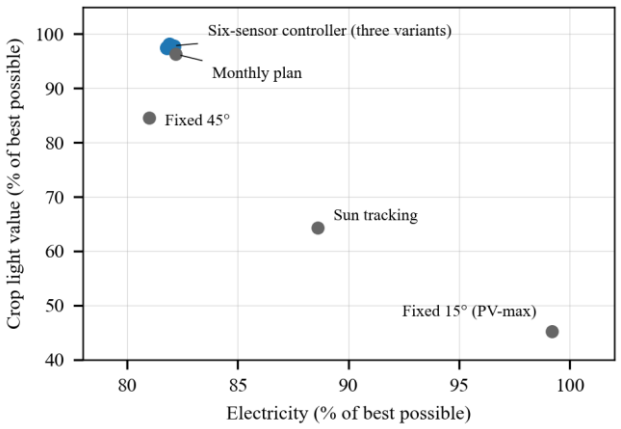


FIGURE 3. ELECTRICITY AND CROP LIGHT. EACH DOT IS A STRATEGY; THE TOP RIGHT IS BEST. THE THREE CONTROLLER VARIANTS AND THE MONTHLY PLAN SIT CLOSE TOGETHER NEAR THE TOP LEFT, THE FIXED 15° PANEL IS FAR RIGHT BUT LOW, AND FIXED 45° AND SUN TRACKING ARE WELL BELOW THE CONTROLLER.

To see where the accuracy comes from, we repeated the test with a different random noise draw and split the cases by sky, soil, and season (Table 4). The new draw gives the same combined score of 99.6 percent (82.1 and 97.7 percent for electricity and crop light in the first draw, 81.9 and 97.8 in the second), so the noise draw moves the scores by about 0.2 points. In every group the combined score is between 99.5 and 99.7 percent, while that of a fixed 45° panel ranges from 89.4 to 95.3 percent. The electricity score depends on the season, from 75.9 percent in October and November to 93.1 percent from March to May, while the combined score stays within 0.2 points. The tilt error is largest under an overcast sky (8.1° against 4.3° in clear sky), possibly because the objective is then nearly flat across angles and a wrong angle costs little. Perfect sensors would raise the combined score only from 99.6 to 99.7 percent, so the sensor noise costs about 0.1 point.

TABLE 4. ACCURACY BY SKY, SOIL, AND SEASON

Group	Energy (%)	Crop (%)	Score (%)	Fixed 45° score (%)
All cases	81.9	97.8	99.6	91.2
Clear sky	82.1	97.4	99.6	89.5
Partly cloudy	81.8	98.9	99.6	94.3
Overcast	81.2	98.1	99.6	95.3
Soil 26 %	81.9	97.8	99.6	91.2
Soil 15 % (dry)	82.0	97.9	99.6	91.3
Dec–Feb	77.8	98.0	99.6	91.7
Mar–May	93.1	99.6	99.7	89.4
Jun–Sep	78.3	96.4	99.5	91.7
Oct–Nov	75.9	97.5	99.6	93.4

Taking the median of five readings and ignoring an implausible PAR reading makes the controller tolerant of noise and faults. At three times the base noise the combined score is 98.5 percent with a single reading and 99.1 percent with the median. With a dirty PAR sensor (reading 0.35 of the true value), a sensor with reflections (1.9 times), or a dead sensor (zero), a single-reading controller scores 98.6, 97.4, and 98.7 percent, while the median-and-guard version scores 99.6 percent in all three. The same test repeated in MATLAB on 142 cases gives 99.5 percent, an electricity score of 83 percent, a crop light score of 99 percent, and a tilt error of 3.4°; the MATLAB and JavaScript codes picked the identical angle in 140 of 142 cases, and the largest difference was 6°.

Seen as a co-optimization, the controller gives up 18 percent of the best possible electricity to keep 98 percent of the best possible crop light value. A fixed 45° panel gives up the same 19 percent of the electricity but keeps only 84.6 percent of the crop light, and a panel fixed for electricity keeps 99 percent of the electricity but only 45 percent of the crop light. In the plane of Fig. 3, the controller sits 2.3 percentage points below the crop-light optimum, and the remaining gap in electricity is the price of that light.

These scores are measured against the controller's own objective. They show that the controller finds the best mix of the two goals; they do not show that crop yield in the field improves by that amount.

YIELD, MONEY AND THE CHOICE OF WEIGHT

To see what the angles are worth, each strategy's hourly angles were run through the engine's crop, water, and economic model for a year. Table 5 gives the values per hectare of canopy per year at assumed prices (tariff INR 3.0 per kWh, farmer share 30 percent, operation and maintenance 1.5 percent of capital cost a year). The controller gives 241.6 kWh/m² of panel and INR 55.74 lakh in total (solar plus crop; 1 lakh is 100,000 rupees), against 238.1 and 55.00 for the monthly plan and 232.6 and 52.46 for a fixed 45° panel. It improves crop income by 36 percent over the fixed 45° panel (INR 2.22 against 1.63 lakh) with 4 percent more electricity.

TABLE 5. YEARLY VALUE PER HECTARE AT ASSUMED PRICES (ENERGY IN KWH/M² OF PANEL, MONEY IN LAKH RUPEES)

Strategy	Energy	Crop net	Solar net	Total
Controller, noisy sensors	241.6	2.22	53.52	55.74
Monthly plan	238.1	2.53	52.47	55.00
Fixed 45°	232.6	1.63	50.83	52.46
Sun tracking	258.0	0.44	58.44	58.88
Fixed 15° (electricity only)	290.7	–0.19	68.23	68.04

Two results go against the controller. The monthly plan earns more from crops (INR 2.53 against 2.22 lakh), and a panel set at 15° for electricity only earns more in total (INR

68.04 lakh) because electricity is worth far more than crops at these prices, while its crop income is slightly negative. We therefore swept the weight w in (4) from zero to one and ran each schedule through the same model (Fig. 4). At $w = 0$ the crop earns INR 2.82 lakh and the total is INR 49.97 lakh; at $w = 1$ the crop earns INR –0.09 lakh and the total is INR 69.22 lakh. At $w = 0.5$ the schedule keeps 82 percent of the largest electricity and 78 percent of the largest crop income.

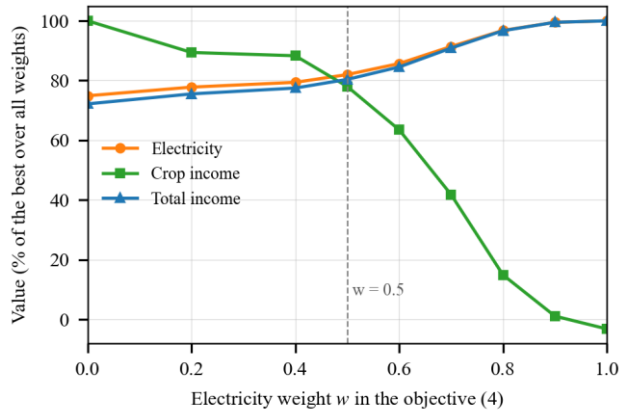


FIGURE 4. EFFECT OF THE ELECTRICITY WEIGHT. ELECTRICITY, CROP INCOME, AND TOTAL INCOME AS PERCENTAGES OF THEIR LARGEST VALUES OVER THE WEIGHTS. TOTAL INCOME FOLLOWS ELECTRICITY CLOSELY, SO A RUPEE OBJECTIVE ALONE WOULD PUSH THE WEIGHT TOWARD ONE AND REMOVE THE CROP.

This is the main limit of the approach. A rupee objective alone would give $w = 1$, and the farm would no longer be a farm. The weight is therefore a design decision, for example a minimum crop income or minimum crop light, and not something the total income can settle. At the weight of 0.5 used here, the controller is a way to keep crop income while losing little electricity compared with a fixed 45° panel.

WATER, STORMS AND WEAR

Water link. The soil probe decides when to irrigate; a fixed timer applies a set depth at a set interval. Table 6 compares them over a year using the same daily balance as the engine. The probe uses 5,372 m³/ha, against 5,687, 5,641, and 5,655 m³/ha for timers every 2, 4, and 7 days, a saving of 4.8 to 5.5 percent. The larger gain is crop water stress: none with the probe, and 14.8, 20, and 29.3 days with the timers. A probe stuck dry without a fallback would apply 46,528 m³/ha, and a probe stuck wet would cut the relative yield from water to 0.155. With the fallback (after three stuck days, or no reading, run the conservative timer schedule), the water use returns to about that of the timer (5,995 and 5,578 m³/ha). The model assumes the borewell supplies water when the cistern is empty, and the probe benefit is against our own timer rules.

TABLE 6. IRRIGATION BY SOIL PROBE AND BY TIMER

Strategy	Water (m³/ha)	Events	Stress days	Yield factor
Soil probe	5,372	99	0	1.000
Timer every 2 days	5,687	331	14.8	0.995
Timer every 4 days	5,641	164	20.0	0.988
Timer every 7 days	5,655	94	29.3	0.966

Storms and rain. We ran the controller for a simulated year of hourly steps with random rain days (monthly rain days from the climate data), generated wind with the monthly mean speed, and four assumed storm days with four hours each at 15 to 24 m/s. These storms are our assumption, not measured wind. With the anemometer, all 16 storm hours are stowed at 90°, and the stow costs about 0.14 percent of the annual electricity. Without it, all 16 storm hours pass with the panels unstowed. The panels are held flat for 222 hours of rain.

Wear. With hourly decisions the panels make about 1,860 moves a year, travel 46,800°, and use 346 m of drive wire, or 918 drum turns. With decisions every five minutes on a clear day, the panels make 64 moves, travel 330°, and use 2.5 m of wire. In a two-hour test of fast cloud flicker the unprotected controller made 92 moves, and the version with a 2 percent improvement threshold made 14.

DIGITAL TWIN AND OTHER SITES

The digital twin is a browser application that runs offline and reproduces the models above for any chosen site. One design file and one data library of climates, soils, and crops feed both the twin and the MATLAB engine. The twin draws the canopy on the college ground in 3D, with building footprints from OpenStreetMap, shows the simulated sensor stack and the controller's decision with its reason at each step, animates the tilt mechanism of Fig. 2, runs the cistern and the soil-probe irrigation, and has tabs for sun and light, the tilt trade-off, rain and water, crops and soil, economics, and accuracy (Fig. 5). Sliders for cloud, temperature, humidity, soil moisture, rain, and wind let a viewer test each rule, and a fault selector injects a faulty rain sensor, soil probe, or light sensor to show the fallbacks. Its sensor values are simulated.



FIGURE 5. DIGITAL TWIN. THE 3D VIEW OF THE CANOPY ON THE COLLEGE GROUND (LEFT) AND THE SIMULATED SENSOR STACK WITH THE CONTROLLER'S DECISION AND ITS REASON (RIGHT), FOR A MORNING IN APRIL.

Rain harvest. With the panels flat, the sealed roof catches rain events of 5 mm or more, after a first flush of 0.5 mm and a runoff coefficient of 0.85, into a 100 m³ cistern. For the checked design about 4,650 m³ per hectare is harvested in a year at Madurai (959 mm of rain) against an irrigation demand of about 4,930 m³. So 94 percent of the irrigation comes from rain, and the borewell tops up about 300 m³. In a drier year (rain 30 percent lower) the rain-fed share is 90.3 percent, and in a wetter year (20 percent more rain) it is 94.4 percent. Sunshine 5 percent lower or higher moves the electricity to 226.6 or 248.6 kWh/m².

Other climates and soils. The same design was run in seven climates and with nine soils, using the four crops above. Table 7 shows that energy ranges from 215 to 243 kWh/m² and mean crop light from 31 to 34 percent, while the rain-fed share ranges from 72 percent in the arid site to 98 percent on the coast. At Madurai, crop net income ranges from INR 3.10 lakh per hectare per year on loam and hill soil to INR 0.39 lakh on saline soil, about eight times lower, so salinity is the main soil risk. Only Madurai uses real building footprints; the other sites use representative landscapes, and the soil and crop values are typical ones.

TABLE 7. THE SAME DESIGN IN SEVEN CLIMATES (ENERGY IN KWH/M² OF PANEL, FARMER INCOME IN LAKH RUPEES PER MONTH PER HECTARE)

Site	Energy	Light (%)	Rain-fed (%)	Farmer
Madurai	237.5	33.7	93.9	1.99
Coastal tropical	225.3	34.3	97.9	1.90
Mountain	226.8	33.8	83.5	1.95
Humid tropical	227.7	33.9	93.0	1.92
Arid desert	242.7	30.9	72.3	1.79
Northern plains	214.7	32.1	82.0	1.74
Deccan plateau	227.9	31.7	75.4	1.80

A Simulink model of the supervisory controller reproduces the rules on test days with cloud, rain, and a wind gust: the panels go flat at 3° in the rain and to 90° in the gust. Together, the MATLAB engine, the twin, and the Simulink model give three implementations of the same rules; the JavaScript and MATLAB controllers choose identical angles in 40 of 40 random cases.

ECONOMICS

At the assumed prices, a Monte Carlo run of 20,000 draws gives farmer income of INR 1.41, 2.12, and 2.99 lakh per month per hectare at the 5th, 50th, and 95th percentiles. The weak link is the investor. The payback time is 21, 37, and 78 years at the same percentiles, the chance of paying back within 20 years is about 3 percent, and a payback of 15 years at a 30 percent farmer share needs a tariff of about INR 6.22 per kWh. Farmer income depends mainly on the contract (tariff and share), not on the crop.

LIMITATIONS

All results come from simulation. The sensors are simulated, the wind and storms are generated, and crop, soil, and price values are typical or assumed. The sky is a monthly mean with an isotropic diffuse part, and the panels are treated as infinite plates. The temperature-only evapotranspiration method is simple and may be less accurate in humid weather. The structural check is screening-level. The closed-loop wire path and the sizes of sheaves and bearing rings are design choices that have not been tested on a bench, and the anchors and foundations need a soil test and an engineer's check. The crop-light scores are measured against the controller's own objective.

CONCLUSION

A six-sensor controller on a wire-driven canopy reaches 99.6 percent of the best possible combined score of electricity and crop light with noisy sensors, and keeps this when a light sensor fails. Against a fixed 45° panel it gives 36 percent more crop income with 4 percent more electricity. A rupee comparison shows that the weight between the two goals cannot be set by income alone, and must be set as a farm design rule. The digital twin reproduces the models for other climates and soils and shows that 94 percent of the irrigation at Madurai can come from harvested rain. The strength check passes for the final design at 140 km/h wind, and in the model the soil-probe water link saves 4.8 to 5.5 percent of the water and avoids crop water stress. The next step is a pilot row with real sensors logging at least one month, followed by supplier quotes, a soil test, and a structural engineer's review.

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ABOUT THE AUTHORS

Author information is withheld for review and will be added to the camera-ready version.